



REFORESTACHAIN

A Blockchain-Based Verification Registry for Reforestation in Peru

Cahuaza Ocmin, Aron Rogelio · *Lead Blockchain Architect & Smart Contract Developer*

Meza Borda, Evelyn Pilar · *Frontend Developer & UI/UX Designer*

Solis Guerrero, Consuelo Felicita · *Environmental & Documentation Analyst*

Vasquez Caceres, Dana Padme · *Video Production & Communications Lead*

Zelada Flores, Fernando José · *QA, Deployment & Roadmap Coordinator*

The Problem: Peru's Deforestation Crisis

A Verification Gap Driven by Negative Externalities (*Pigou, 1920*). The party that clears or degrades forest rarely bears the full social cost of the resulting carbon emissions and biodiversity loss — and the reforestation that does happen is difficult for outsiders to confirm.

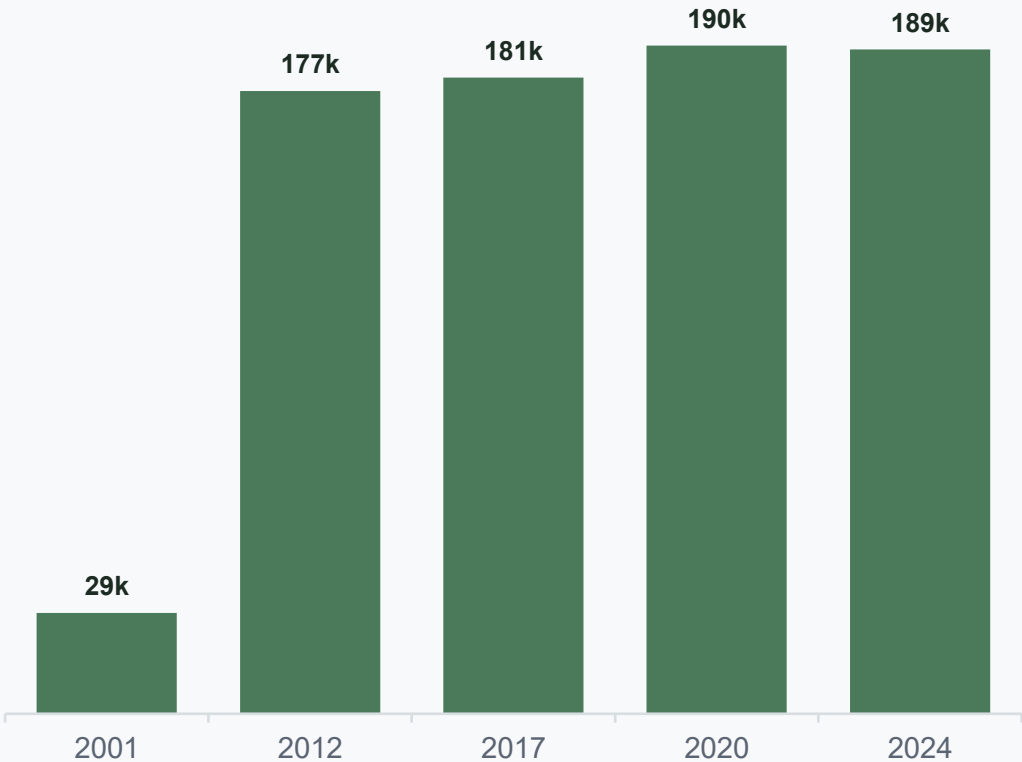
189,005 ha

Hectares of primary forest lost in 2024 alone

3.2M ha

Cumulative Amazon forest loss, 2001–2024

Peru: Annual Primary Forest Loss (2001–2024)



Source: Ceplan (2025), based on Global Forest Watch.

Structural Frictions: Why Claims Are Hard to Trust



Transaction Costs

(Coase, 1960)

Confirming that trees were actually planted, and that they survived, typically requires field visits, drone flights, or satellite analysis — costs most small reforestation projects cannot afford.



Information Asymmetry

(Akerlof, 1970)

Donors, carbon-credit buyers, and government agencies cannot easily distinguish a well-executed planting project from an unverified or overstated one, which discourages funding even for genuine initiatives.

The Solution: ReforestaChain

An EVM-compatible smart contract deployed on the LACNet blockchain that issues an immutable “Proof of Planting” record whenever a reforestation event is confirmed by an authorized validator.



Field Registration

A community member or partner organization plants trees and records location, species, and quantity.

Verified Confirmation

An authorized validator — partner NGO, community leader, or forestry office — confirms the event via `registerPlanting()`.

On-Chain Certification

The contract issues a non-transferable “Proof of Planting” record tied to the executing organization’s wallet.

Every certificate permanently links a reforestation claim to verifiable, tamper-proof on-chain evidence.

Technical Rigor: Architecture & Gas Optimization



Solidity & EVM

A transparent, immutable smart contract core deployed on the LACNet Pro-Testnet, removing the need for a costly third-party auditor.



O(1) Gas Optimization

Mapping-based storage for constant-time lookups, minimizing on-chain state writes to respect LACNet's per-block gas limit.



Access Controls

Lightweight modifiers restrict certificate issuance to authorized validators, avoiding redundant on-chain checks.



```
// ReforestaChain.sol - excerpt  
function registerPlanting(  
    address executor, uint256 qty,  
    string calldata species, string calldata geo  
) external onlyValidator {  
    require(qty > 0, "Invalid quantity");  
    plantings[executor].push(  
        Proof(qty, species, geo,  
            block.timestamp)  
    );  
    emit PlantingVerified(executor, qty, geo);  
}
```

Frontend & UX: Field-Facing Web3 Interface



MetaMask Authentication

Direct wallet integration standardizes access for field operators and partner organizations — no centralized passwords.



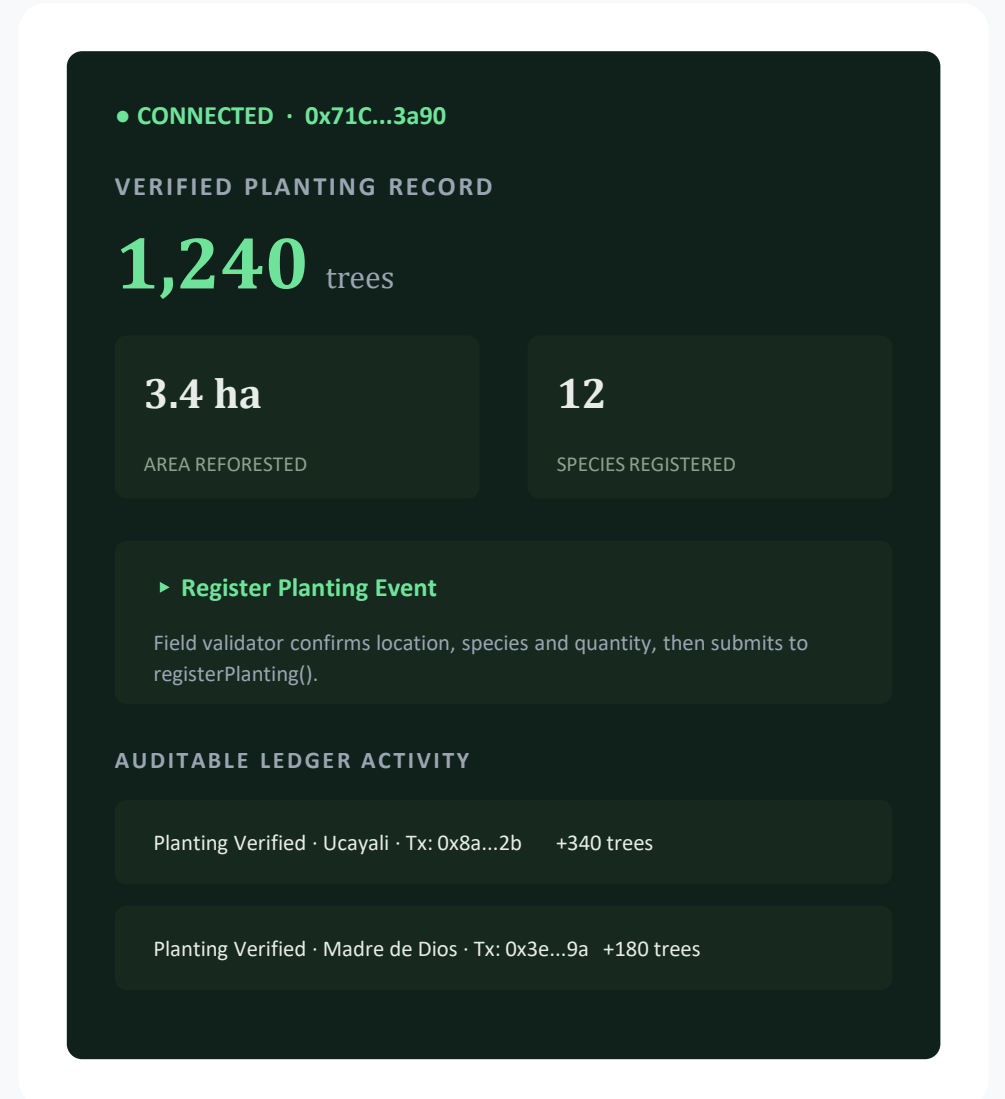
Registration Dashboard

React + ethers.js interface for registering planting events and looking up on-chain certificates in real time.

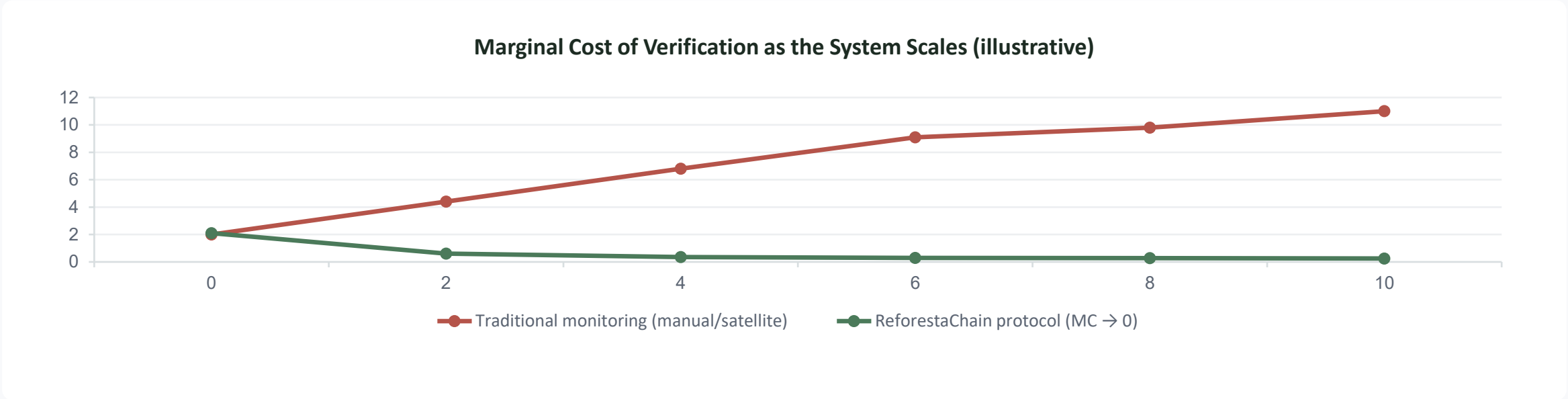


Field Validator View

A simple, low-friction flow lets authorized validators confirm plantings from the field in a single transaction.



Impact & Scalability



Marginal Cost Tending to Zero (MC → 0)

Zero-cost node participation and no added physical infrastructure per site mean onboarding a new organization or region costs almost nothing.



Signaling Theory (Spence, 1973)

An immutable, validator-confirmed record is a costly, hard-to-fake signal, letting funders verify genuine performance without repeated audits.

Strategic Roadmap: Post-Hackathon Projection

Q3 2026



Technical Rigor & Security

Conduct a vulnerability review of the Solidity code and refine gas optimization. Success metric: zero critical vulnerabilities, reduced average gas cost.

Q4 2026



Field Pilot Validation

Launch a small pilot with one reforestation site or partner community, testing registerPlanting() with a real field validator.

Q1 2027



Ecosystem Scalability

Integrate with regional forestry offices and open an aggregated, anonymized dashboard of verified planting activity.